CS215 OPEN PROJECT

Does poverty track health across India’s districts?

District-level multidimensional poverty (MPI) and eight NFHS-5 health indicators

Research question How is district-level MPI associated with selected health indicators?

707

districts, one row each

8

health indicators, all in percent

2019-21

NFHS-5 survey period

Group 8. Rishabh Meel, Sugam Kumar Thakur, Aaryan Chavda, Arshdeep Singh

Two sources, one clean table of 707 districts

1

Load

Health fact sheet: 707 districts. MPI table: 709 districts.

2

Select

District, state, MPI and the 8 health indicators.

3

Merge

Match on state and district: 614 exact, 30 alias, 47 fuzzy, 16 manual.

4

Check

No duplicates. Missing: 22 in vaccination, 1 in adequate diet. Not imputed.

5

Analyse

Pearson and Spearman correlations, plus simple linear regression.

707

districts after merge

8

health indicators

0

duplicate districts

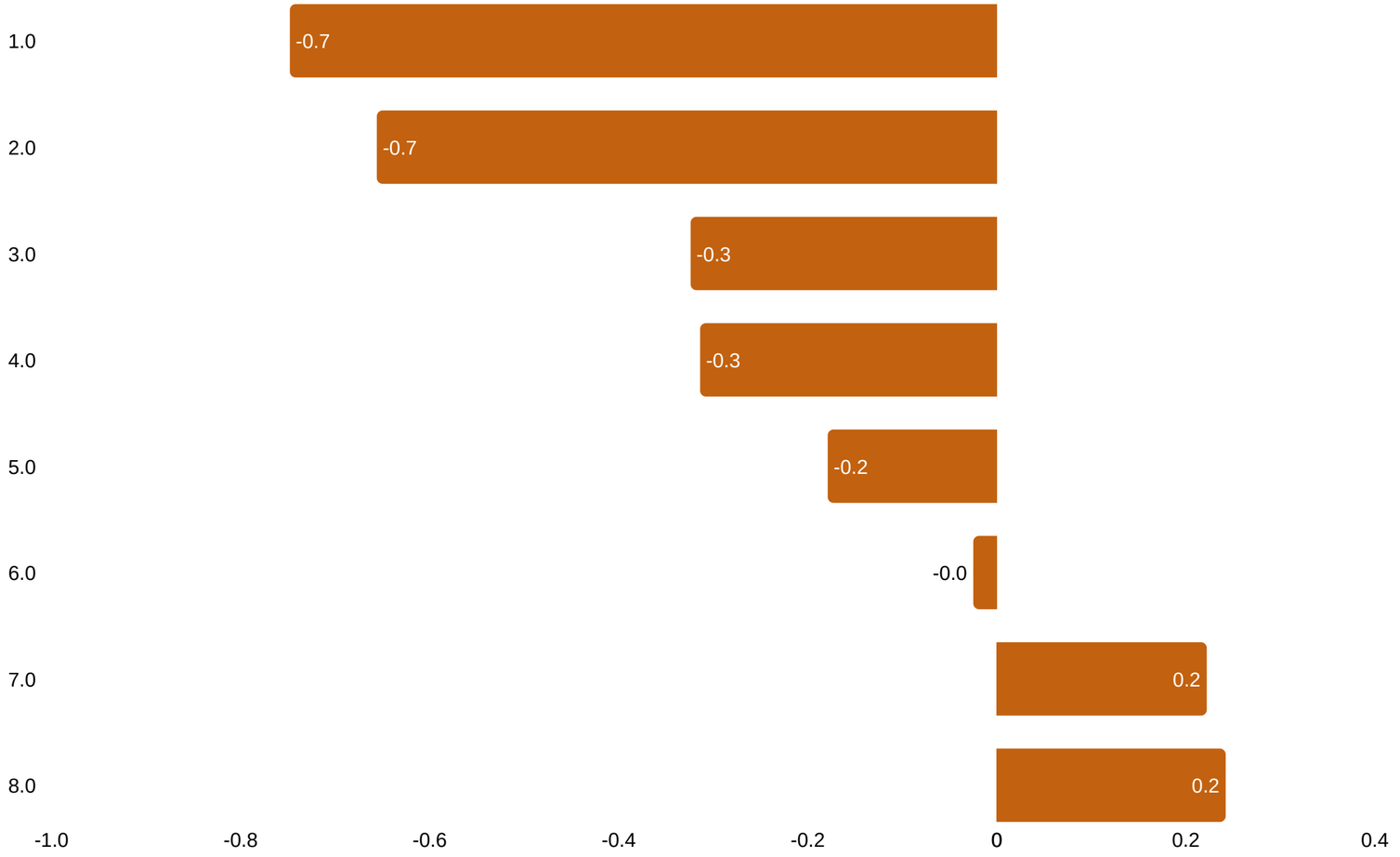
23

missing values, in 2 indicators

Source: NFHS-5 (2019-21) district fact sheet and district MPI table. Missing values are left as missing, not imputed.

Hygiene and sanitation track poverty most closely

Pearson correlation of each indicator with MPI, 8 indicators, 707 districts



−0.748

Menstrual hygiene, the strongest link

−0.656

Improved sanitation

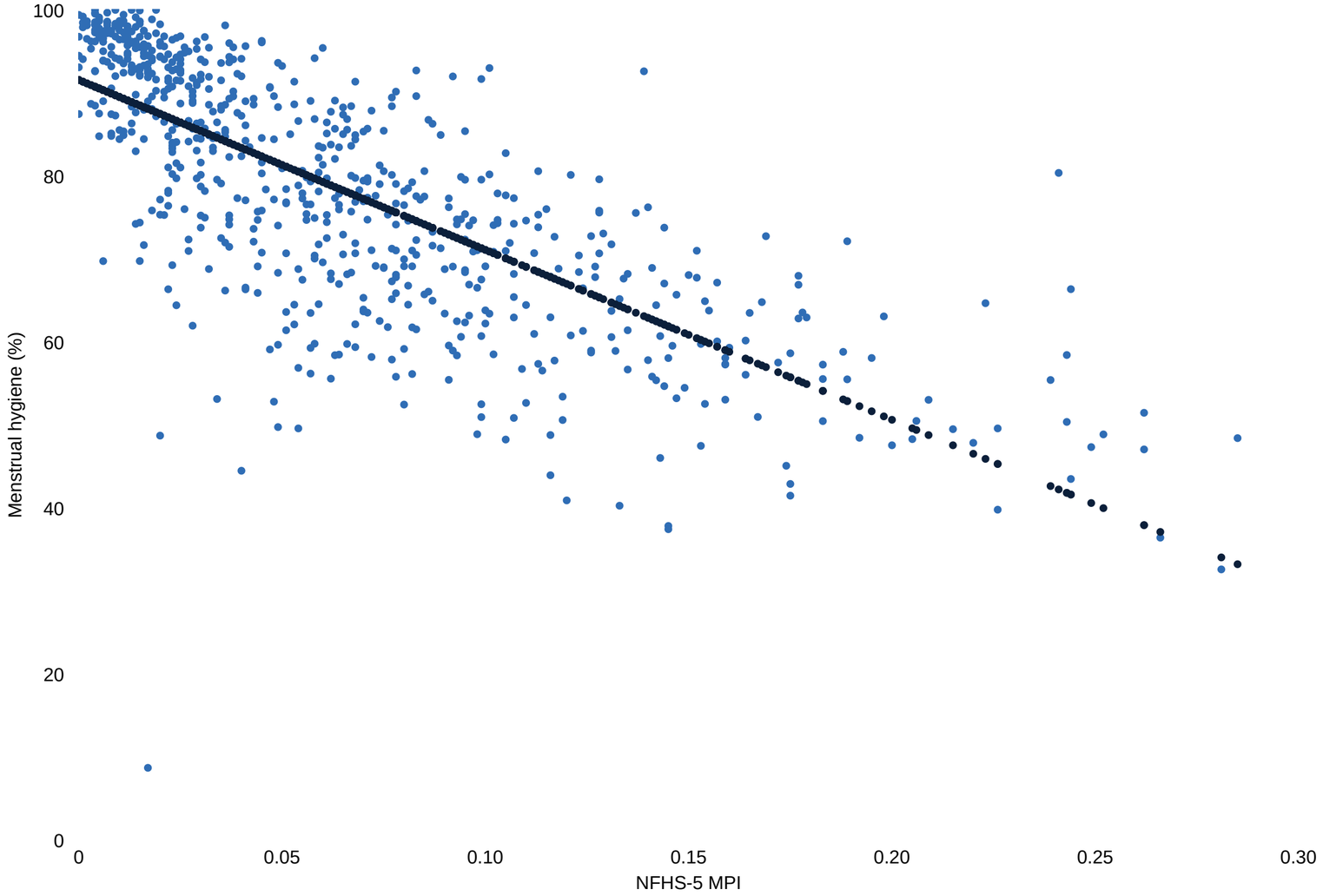
How to read it

Blue bars: coverage is lower where MPI is higher. Orange bars: the indicator is higher where MPI is higher.

Source: NFHS-5 (2019-21) district fact sheet and district MPI table. Observational data, correlation is not causation.

Menstrual hygiene falls about 20 points per 0.1 MPI

Menstrual hygiene (%) against district MPI, one dot per district (n = 707)



−0.748
Pearson r

−0.795
Spearman ρ

0.559
R^2, share of variation explained

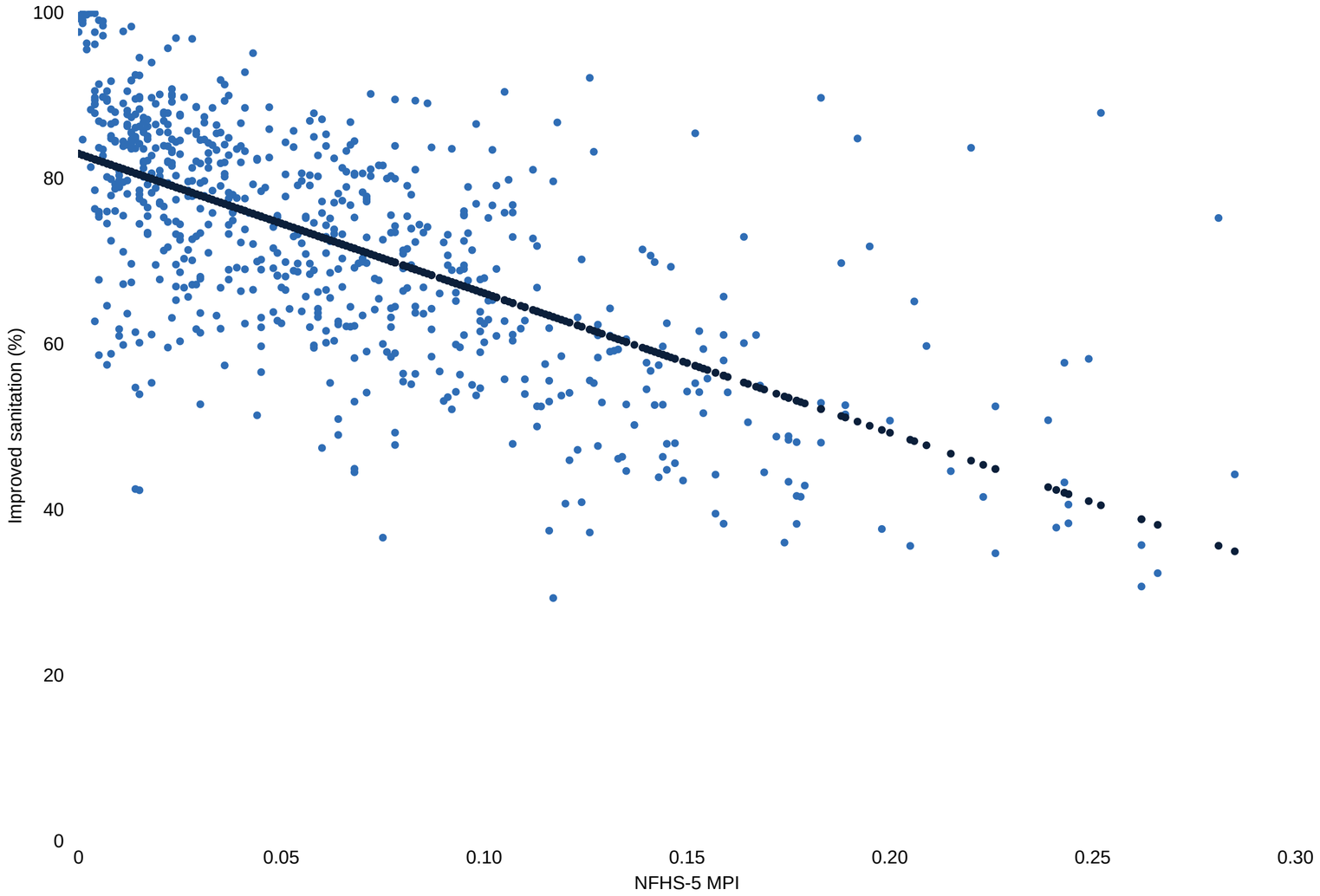
−20.5
percentage points per +0.1 MPI

Descriptive, not causal. MPI alone leaves about 44 percent of the variation unexplained.

Dark line is the unadjusted linear fit. Slope -204.6 per 1.0 MPI, $p < 0.001$. Source: NFHS-5 (2019-21) and district MPI table.

Sanitation follows the same pattern, with a looser fit

Improved sanitation (%) against district MPI, one dot per district (n = 707)



−0.656
Pearson r

−0.640
Spearman ρ

0.430
R^2, share of variation explained

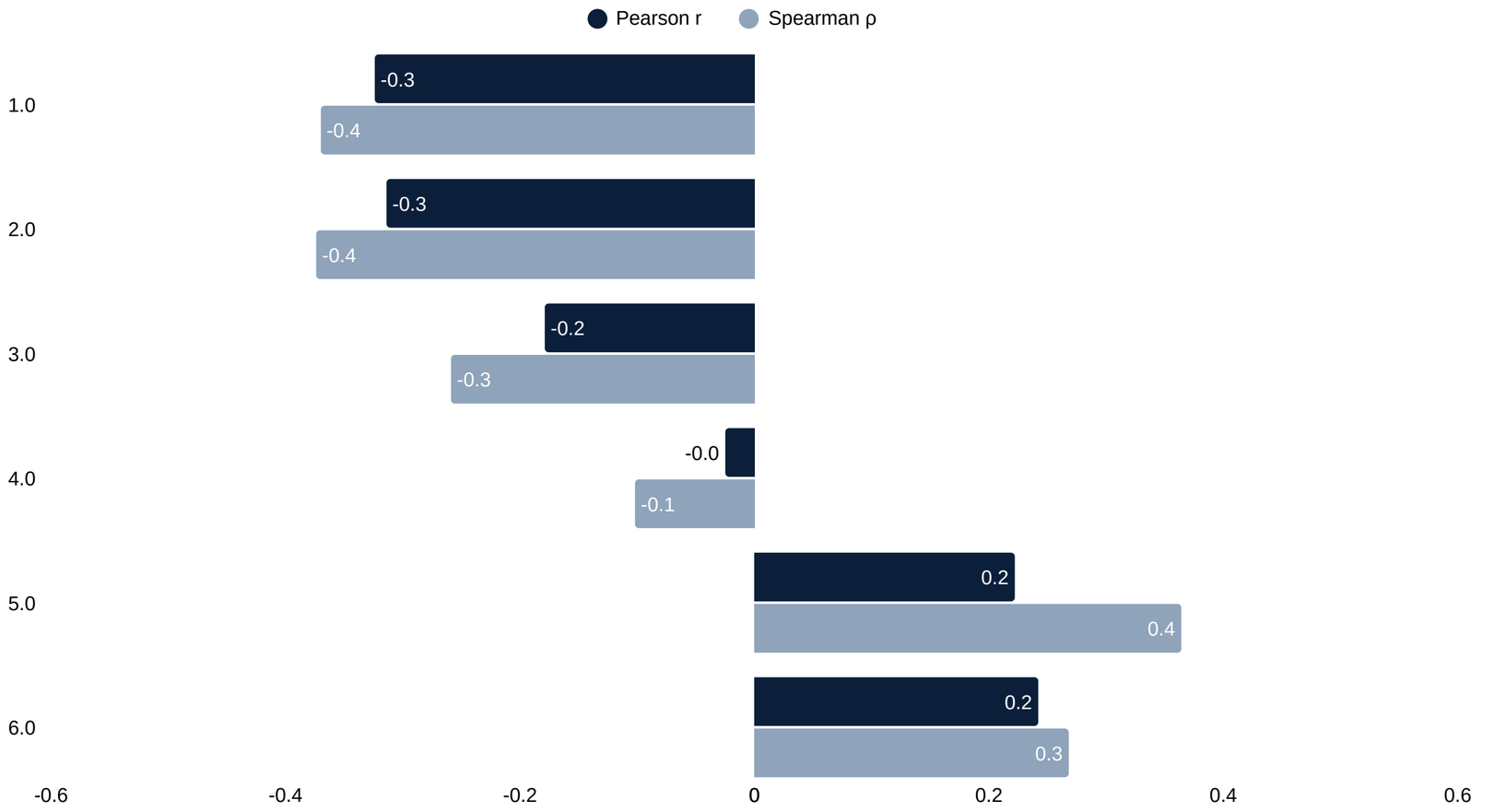
−16.8
percentage points per +0.1 MPI

Descriptive, not causal. Sanitation is also a component of MPI, so part of this link is built in.

Dark line is the unadjusted linear fit. Slope -168.3 per 1.0 MPI, $p < 0.001$. Source: NFHS-5 (2019-21) and district MPI table.

The other six indicators are weaker or mixed

Pearson r and Spearman ρ with MPI. Rank-based and linear measures usually agree



Anaemia and tobacco rise

Anaemia in women (r +0.242) and tobacco use among women (r +0.222) are positive.

Tobacco: rank beats linear

ρ +0.364 is much larger than r. Suggests a non-linear pattern or influential districts.

Alcohol: no relationship

r −0.025, p = 0.51, not statistically significant.

Lower blood sugar in higher-MPI districts likely reflects screening or age mix, not a benefit of poverty. Source: NFHS-5 (2019-21).

A strong association, with limited inference

Higher-poverty districts tend to report lower hygiene and sanitation coverage.

Limitations

- District averages can hide variation within districts
- Unadjusted correlations, no controls
- Some missing data, in vaccination and diet
- Manual name matching across two files
- No population weighting
- Sanitation is also part of MPI, so overlap is built in

Observational, cross-sectional district data. Correlation is not causation.

Thank you

Higher-poverty districts tend to report lower hygiene and sanitation coverage. Associations, not causes.